

Figure 1: Branches of GitHub

Listed below are some GitHub commands:

- `git status`
- `git add <files with proper path>`
- `git commit -m "give proper message"`
- `git push origin <your local branch>`
- `git checkout development`
- `git merge <local branch name>`
- `git add <files with proper path>`
- `git commit -m "give proper message"`
- `git push origin main repository`

Build tools

Build tools automate and orchestrate the process of compiling source code, running tests, downloading dependencies (JAR files), creating a package and deploying packages. Earlier, the process was manual and now, build tools have made the automation process easy. Build tools are an integral part of the DevOps practices such as continuous integration. Ant, Maven and Gradle are build tools for Java; NAnt is used for the .NET framework and MsBuild is used for C#.

Gradle

Gradle is an open source build automation system that builds upon the concept of Apache Ant and Apache Maven, and introduces a Groovy based domain-specific language instead of the XML form used by Apache Maven for declaring the project configuration.

How to install Gradle in your Windows system: There are two ways to set up Gradle in Windows.

To install it manually:

- Go to the Gradle official website: <https://gradle.org/>.
- Download the latest version of the Gradle distribution file.
- Unzip it in your program file folder or your specific path.
- Now set the Gradle path in your system's environment variable.
- Go to *System Variable* and add a new path.
- Click *Okay*.
- Now add the path `C:\Program Files\gradle-4.4.1\bin`.
- Click on *Okay*. Go to the command prompt and check the Gradle version by using the `gradle -v` command.

Installing Gradle on Android Studio:

- Go to Android Studio.
- Open your existing project in Android Studio.

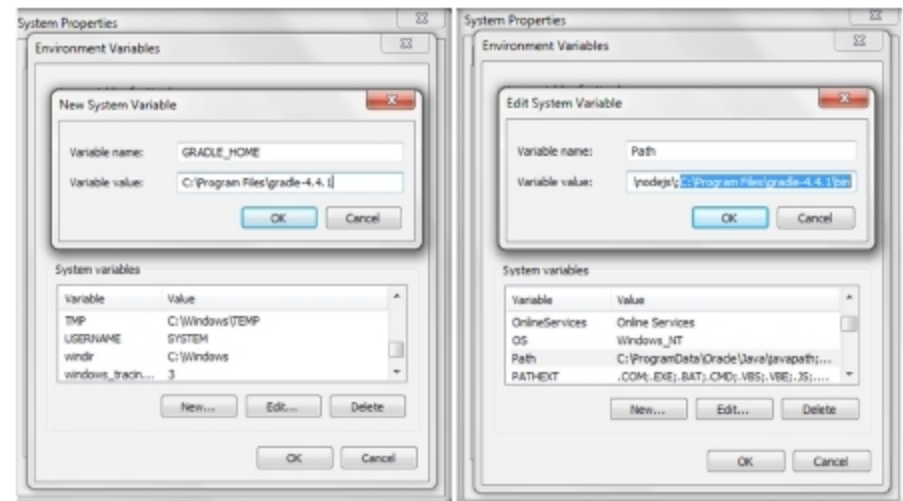


Figure 2: Environment variables

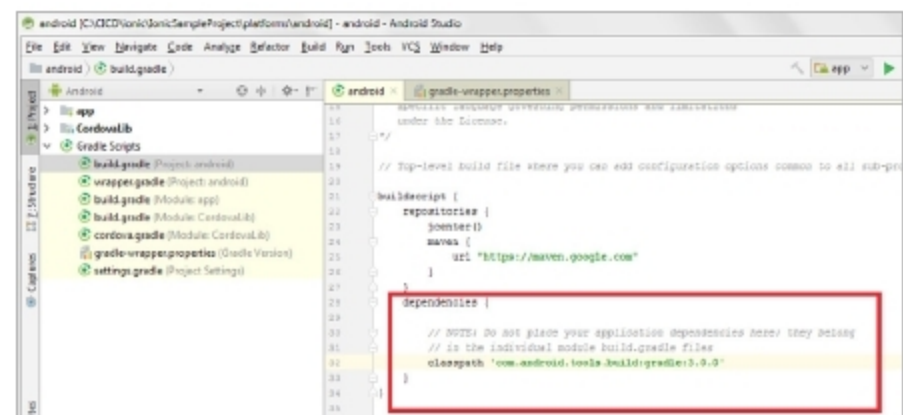


Figure 3: Android Studio Dependencies

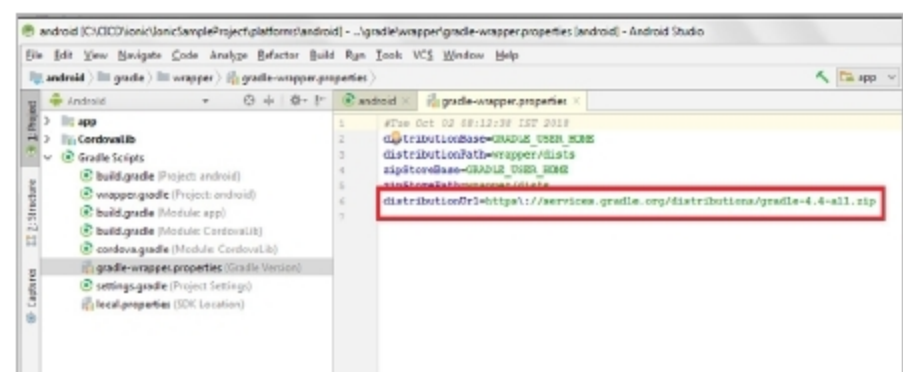


Figure 4: Properties of gradle wrapper

- Go to the *build.gradle* file for your project in the project module.
- Inside *Dependencies*, look for the *classpath* of Gradle.
- Go to the *gradle-wrapper.properties* file.
- Give the distribution URL for online Gradle installation, specifying the particular version.

Stable release	4.10.2
Licence	Apache License
Website	https://gradle.org/
Written in	Java, Groovy, Kotlin
On-premise or hosted	Package available.
Main features	Groovy-based domain-specific language (DSL). Works well with multi-project builds. Supports Android, Java, and so on.